

ESM3. Additional Results

TABLE I: Recall Test Results. The **best result** within Protocol A is **bolded**.

Model Configuration	Recall – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.6648 (0.0137)	0.6489 (0.0291)	0.3147 (0.0232)	0.3595 (0.0325)
Frozen + Attention	0.7792 (0.0140)	0.7880 (0.0192)	0.4463 (0.0219)	0.4978 (0.0028)
Fine-tuned + Last Layer + No Attention	0.6658 (0.0280)	0.6605 (0.0141)	0.3492 (0.0407)	0.3855 (0.0121)
Fine-tuned + Last Layer + Attention	0.7966 (0.0072)	0.7957 (0.0155)	0.4422 (0.0345)	0.5327 (0.0228)
Fine-tuned + Last 5 Layers + No Attention	0.7371 (0.0177)	0.7090 (0.0335)	0.6328 (0.0146)	0.6366 (0.0147)
Fine-tuned + Last 5 Layers + Attention	0.8472 (0.0085)	0.8557 (0.0113)	0.7195 (0.0157)	0.7064 (0.0089)
Fine-tuned + Last 10 Layers + No Attention	0.8740 (0.0069)	0.8101 (0.0152)	0.8453 (0.0129)	0.6670 (0.0393)
Fine-tuned + Last 10 Layers + Attention	0.8757 (0.0200)	0.8511 (0.0057)	0.8691 (0.0147)	0.7524 (0.0181)
Fine-tuned + Last 25 Layers + No Attention	0.9069 (0.0070)	0.8362 (0.0240)	0.8665 (0.0204)	0.8305 (0.0108)
Fine-tuned + Last 25 Layers + Attention	0.9178 (0.0100)	0.8703 (0.0217)	0.8738 (0.0146)	0.8292 (0.0004)
Fine-tuned + No Attention	0.9348 (0.0232)	0.9561 (0.0175)	0.9727 (0.0039)	0.9595 (0.0094)
Fine-tuned + Attention	0.9448 (0.0190)	0.9730 (0.0033)	0.9697 (0.0041)	0.9703 (0.0051)
From Scratch + No Attention	0.8828 (0.0345)	0.9042 (0.0121)	0.9322 (0.0040)	0.1418 (0.0967)
From Scratch + Attention	0.8848 (0.0310)	0.9030 (0.0231)	0.9438 (0.0015)	0.0056 (0.0000)

TABLE II: Precision Test Results. The **best result** within Protocol A is **bolded**.

Model Configuration	Precision – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.7387 (0.0155)	0.7128 (0.0236)	0.3645 (0.0251)	0.4479 (0.0453)
Frozen + Attention	0.8146 (0.0183)	0.8199 (0.0257)	0.5017 (0.0236)	0.5850 (0.0068)
Fine-tuned + Last Layer + No Attention	0.7265 (0.0254)	0.7293 (0.0106)	0.4017 (0.0464)	0.4800 (0.0068)
Fine-tuned + Last Layer + Attention	0.8350 (0.0086)	0.8352 (0.0085)	0.5025 (0.0449)	0.6166 (0.0192)
Fine-tuned + Last 5 Layers + No Attention	0.7867 (0.0281)	0.7759 (0.0264)	0.7032 (0.0072)	0.7107 (0.0179)
Fine-tuned + Last 5 Layers + Attention	0.8811 (0.0159)	0.8819 (0.0131)	0.7602 (0.0152)	0.7661 (0.0051)
Fine-tuned + Last 10 Layers + No Attention	0.8984 (0.0104)	0.8400 (0.0207)	0.8781 (0.0151)	0.7339 (0.0428)
Fine-tuned + Last 10 Layers + Attention	0.9025 (0.0093)	0.8753 (0.0029)	0.8977 (0.0099)	0.8129 (0.0166)
Fine-tuned + Last 25 Layers + No Attention	0.9233 (0.0027)	0.8659 (0.0135)	0.8948 (0.0274)	0.8526 (0.0189)
Fine-tuned + Last 25 Layers + Attention	0.9352 (0.0087)	0.8897 (0.0193)	0.8990 (0.0205)	0.8463 (0.0040)
Fine-tuned + No Attention	0.9378 (0.0294)	0.9638 (0.0168)	0.9739 (0.0036)	0.9631 (0.0094)
Fine-tuned + Attention	0.9500 (0.0213)	0.9819 (0.0032)	0.9803 (0.0010)	0.9765 (0.0038)
From Scratch + No Attention	0.9006 (0.0323)	0.9242 (0.0120)	0.9503 (0.0053)	0.1141 (0.0808)
From Scratch + Attention	0.9058 (0.0327)	0.9298 (0.0125)	0.9606 (0.0041)	0.0003 (0.0000)

TABLE III: Test Loss Results. The **best result** within Protocol A is **bolded**.

Model Configuration	Test Loss – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.5517 (0.0088)	0.5415 (0.0235)	1.2822 (0.0502)	1.1310 (0.0181)
Frozen + Attention	0.3995 (0.0145)	0.3747 (0.0138)	1.0810 (0.0227)	0.9310 (0.0429)
Fine-tuned + Last Layer + No Attention	0.5230 (0.0072)	0.5379 (0.0225)	1.2475 (0.0528)	1.0789 (0.0143)
Fine-tuned + Last Layer + Attention	0.3998 (0.0178)	0.3735 (0.0155)	1.1140 (0.0109)	0.8551 (0.0403)
Fine-tuned + Last 5 Layers + No Attention	0.4553 (0.0024)	0.4321 (0.0246)	0.7992 (0.0418)	0.6212 (0.0149)
Fine-tuned + Last 5 Layers + Attention	0.3364 (0.0196)	0.3125 (0.0309)	0.7460 (0.0176)	0.5684 (0.0215)
Fine-tuned + Last 10 Layers + No Attention	0.3037 (0.0072)	0.3056 (0.0097)	0.3988 (0.0433)	0.5769 (0.0171)
Fine-tuned + Last 10 Layers + Attention	0.3377 (0.0294)	0.2786 (0.0150)	0.3691 (0.0195)	0.5454 (0.0218)
Fine-tuned + Last 25 Layers + No Attention	0.2779 (0.0063)	0.3093 (0.0227)	0.3443 (0.0130)	0.5018 (0.0452)
Fine-tuned + Last 25 Layers + Attention	0.2489 (0.0288)	0.2784 (0.0264)	0.3497 (0.0163)	0.5051 (0.0342)
Fine-tuned + No Attention	0.1161 (0.0147)	0.0876 (0.0135)	0.0958 (0.0134)	0.1046 (0.0186)
Fine-tuned + Attention	0.1052 (0.0226)	0.0694 (0.0061)	0.0902 (0.0035)	0.0947 (0.0067)
From Scratch + No Attention	0.1597 (0.0132)	0.1455 (0.0056)	0.1119 (0.0093)	2.2607 (1.5662)
From Scratch + Attention	0.1613 (0.0231)	0.1600 (0.0123)	0.1386 (0.0068)	4.6212 (0.1239)

TABLE IV: Recall Test Results. The **best result** within Protocol B is **bolded**.

Model Configuration	Recall – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.5949 (0.0183)	0.5530 (0.0199)	0.3827 (0.0295)	0.4839 (0.0078)
Frozen + Attention	0.6046 (0.0289)	0.6194 (0.0388)	0.4456 (0.0471)	0.5228 (0.0401)
Fine-tuned + Last Layer + No Attention	0.5679 (0.0164)	0.5578 (0.0181)	0.3752 (0.0407)	0.4782 (0.0409)
Fine-tuned + Last Layer + Attention	0.6180 (0.0251)	0.6089 (0.0510)	0.4265 (0.0373)	0.5234 (0.0417)
Fine-tuned + Last 5 Layers + No Attention	0.6128 (0.0193)	0.5273 (0.0809)	0.5334 (0.0433)	0.5592 (0.0325)
Fine-tuned + Last 5 Layers + Attention	0.6235 (0.0275)	0.5420 (0.1681)	0.5248 (0.0877)	0.5611 (0.1007)
Fine-tuned + Last 10 Layers + No Attention	0.6345 (0.0278)	0.6034 (0.0667)	0.6438 (0.0181)	0.5412 (0.0829)
Fine-tuned + Last 10 Layers + Attention	0.6388 (0.0196)	0.5794 (0.1089)	0.6164 (0.0549)	0.5381 (0.0889)
Fine-tuned + Last 25 Layers + No Attention	0.6783 (0.0155)	0.6383 (0.0486)	0.6780 (0.0031)	0.6179 (0.0311)
Fine-tuned + Last 25 Layers + Attention	0.6787 (0.0144)	0.6170 (0.0758)	0.6688 (0.0107)	0.6445 (0.0266)
Fine-tuned + No Attention	0.6438 (0.0138)	0.6333 (0.0102)	0.6816 (0.0282)	0.6885 (0.0108)
Fine-tuned + Attention	0.6333 (0.0294)	0.6832 (0.0089)	0.6777 (0.0125)	0.6865 (0.0135)
From Scratch + No Attention	0.5381 (0.0435)	0.5438 (0.0224)	0.5780 (0.0926)	0.0141 (0.0000)
From Scratch + Attention	0.5567 (0.0381)	0.5376 (0.0290)	0.5627 (0.0954)	0.0141 (0.0000)

TABLE V: Precision Test Results. The **best result** within Protocol B is **bolded**.

Model Configuration	Precision – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.5484 (0.0312)	0.5139 (0.0403)	0.3355 (0.0205)	0.4606 (0.0239)
Frozen + Attention	0.5700 (0.0307)	0.5796 (0.0238)	0.4094 (0.0334)	0.4984 (0.0421)
Fine-tuned + Last Layer + No Attention	0.5267 (0.0387)	0.5165 (0.0089)	0.3325 (0.0476)	0.4482 (0.0492)
Fine-tuned + Last Layer + Attention	0.5867 (0.0355)	0.5695 (0.0521)	0.3720 (0.0344)	0.5021 (0.0465)
Fine-tuned + Last 5 Layers + No Attention	0.5761 (0.0281)	0.4895 (0.0994)	0.4812 (0.0459)	0.5077 (0.0322)
Fine-tuned + Last 5 Layers + Attention	0.5789 (0.0364)	0.4870 (0.1801)	0.4719 (0.0900)	0.5024 (0.1058)
Fine-tuned + Last 10 Layers + No Attention	0.5841 (0.0330)	0.5558 (0.0698)	0.6148 (0.0124)	0.4855 (0.0924)
Fine-tuned + Last 10 Layers + Attention	0.5788 (0.0256)	0.5140 (0.1259)	0.5627 (0.0670)	0.4808 (0.0917)
Fine-tuned + Last 25 Layers + No Attention	0.6244 (0.0124)	0.5759 (0.0623)	0.6422 (0.0076)	0.5670 (0.0470)
Fine-tuned + Last 25 Layers + Attention	0.6474 (0.0105)	0.5611 (0.0941)	0.6332 (0.0070)	0.5979 (0.0142)
Fine-tuned + No Attention	0.5754 (0.0402)	0.5694 (0.0432)	0.6347 (0.0145)	0.6238 (0.0291)
Fine-tuned + Attention	0.5726 (0.0590)	0.6235 (0.0358)	0.6142 (0.0198)	0.6460 (0.0065)
From Scratch + No Attention	0.4218 (0.0719)	0.4452 (0.0290)	0.4974 (0.1076)	0.0008 (0.0007)
From Scratch + Attention	0.4747 (0.0705)	0.4425 (0.0484)	0.4644 (0.1298)	0.0002 (0.0001)

TABLE VI: Test Loss Results. The **best result** within Protocol B is **bolded**.

Model Configuration	Test Loss – Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	1.5836 (0.0897)	1.6863 (0.0748)	2.4914 (0.0505)	2.0683 (0.1450)
Frozen + Attention	1.6172 (0.1343)	1.7234 (0.0900)	2.4198 (0.0357)	2.1090 (0.1062)
Fine-tuned + Last Layer + No Attention	1.6338 (0.0890)	1.6851 (0.1179)	2.5855 (0.0939)	2.0822 (0.1324)
Fine-tuned + Last Layer + Attention	1.6498 (0.1410)	1.6924 (0.0744)	2.4164 (0.0172)	2.0582 (0.1078)
Fine-tuned + Last 5 Layers + No Attention	1.7123 (0.1504)	1.7525 (0.0914)	2.2157 (0.0450)	1.9285 (0.1250)
Fine-tuned + Last 5 Layers + Attention	1.6055 (0.0387)	1.9828 (0.3119)	2.2562 (0.0729)	2.1362 (0.0703)
Fine-tuned + Last 10 Layers + No Attention	1.8789 (0.1659)	1.7603 (0.1170)	2.1344 (0.2775)	2.1522 (0.2272)
Fine-tuned + Last 10 Layers + Attention	1.7746 (0.1674)	1.8089 (0.1878)	2.0095 (0.1133)	2.0209 (0.0517)
Fine-tuned + Last 25 Layers + No Attention	1.9069 (0.2826)	1.6937 (0.1449)	2.0380 (0.2488)	2.1044 (0.3570)
Fine-tuned + Last 25 Layers + Attention	1.7816 (0.1589)	1.7279 (0.1172)	1.9592 (0.1395)	2.2494 (0.2569)
Fine-tuned + No Attention	1.7742 (0.0613)	1.7317 (0.1221)	2.0275 (0.0528)	1.5808 (0.1075)
Fine-tuned + Attention	1.7843 (0.1066)	1.7852 (0.1248)	1.9128 (0.2054)	1.7630 (0.0797)
From Scratch + No Attention	1.9663 (0.1899)	1.9407 (0.1358)	1.9831 (0.1813)	4.3754 (0.0656)
From Scratch + Attention	2.1376 (0.2081)	2.0616 (0.1529)	2.2160 (0.0523)	4.4023 (0.1125)